3DUptake User Manual

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1. Introduction

Hardware requirements

Compatible with Windows/Linux/MacOS

- CUDA is recommended to accelerate Cellpose segmentation.
- Apple silicon GPU supported with MPS.

Acquisition requirements

Confocal microscopy – high-speed oriented microscope is recommended.

For more information check Key considerations.

Required software:

FIJI/ImageJ

Cellpose

Python environment (miniforge, Anaconda, or python venv)

BIOP Wrapper for Fiji

Tested versions

Fiji 1.54p and later

Cellpose v2 – latest (v4.2)

Python 3.9 - 3.12

2. Experimental workflow

Cell setup protocol

Protocol can be found on Protocol.io linked here.

Cell setup

1 Seed cells in Imaging-compatible tissue culture dishes in appropriate medium at [M] 0.1 M/mL for Overnight at 37˚C and 5% CO_2.
Examples dishes:
35mm glass-bottom dish
Glass-bottom chambered slides
Glass-bottom imaging plate

1.1 Example cell culture:
RAW264.7 (**ATCC**) macrophages are seeded to 35mm glass-bottom MatTek dishes at [M] 0.1 M/mL in RPMI 1640 Medium **Thermo Fisher Catalog #21875059** with 10% Fetal Bovine Serum, qualified, heat inactivated, Australia **Thermo Fisher Catalog #10100147** and 1% L-Glutamine (200 mM) **Thermo Fisher Catalog #25030164** for Overnight .

Imaging Acqusition

2 Set confocal microscope chamber to 37˚C and 5% CO_2, and preheat culture medium to 37˚C.

3 Set up imaging with an inverted confocal microscope using the 60x/63x oil objective, select the channels required and find focus.

4 Mix fluorescent cargo to desired concentration in medium. Keep the mixture warm.
- If cargo is dissolved in DMSO, the final DMSO concentration must be below 1% v/v.

4.1 Prepare the medium/fluorescent cargo mixture. To conserve fluorescent cargo, minimum 80µL of medium required for 35mm glass-bottom dish with 14mm glass, or 100µL of medium for 8-chambered slide.

5 While the dish is on the microscope stage, remove all the medium carefully, then quickly replace with the medium/fluorescent cargo mixture. Make sure to not bump/move the dish

- In 35mm glass bottom dish, pipette directly onto the glass portion of the dish

5.1 Adjust any parameters as needed.

6 Capture 3D Z-stacks at random positions on the dish within 20mins of incubation.

6.1 Choose the positions randomly, and minimise partial inclusion of the cell on the edge of the field of view.

7 After acquisition of images is completed, make sure that the data is transferred/backed up where it can be easily accessed for image analysis.

Imaging Analysis

8 Ensure that the analysis is being carried out on a system with the appropriate computing power.

Setup FIJI, BIOP, and cellpose for image processing. Installation guide and FIJI batch script can be found **Here**.

8.1 Parameters need to be optimised for specific cell type. Optimal threshold method and target vesicle size can be tested with example image in FIJI. Optimal cell diameter, cell probability, and flow threshold can be tested in cellpose GUI. Guide on how to use cellpose GUI can be found **Here**.

8.2 To fast-track image processing, the image can be converted from proprietary imaging format to .tiff. Batch conversion script can be found **Here**.

9 Import 3DUptake.ijm using FIJI. Detailed instructions on how to import the script can be found **Here.**

10 When prompted, choose the correct imaging format, input directory, and optimised parameters, then click 'OK'

10.1 The Log window will update on the status of the batch processing, wait for the process to finish

11 The output quantification can be found under /*Input folder*/Output.
The output folder contains .csv file for all measurements, and a reference folder, which contains the transcript of the log window, and summary of segmentated image.
- Channel 1: cell body
- Channel 2: isolated vesicles
- Channel 3: Cellpose segmentation

3. Installation

Install python environment

miniforge

https://github.com/conda-forge/miniforge

alternatively, install:

Anaconda

Download link and installation instructions can be found on Anaconda official website: cre

(Local admin access to the computer is required to install Anaconda)

Install Fiji

FIJI download and installation:

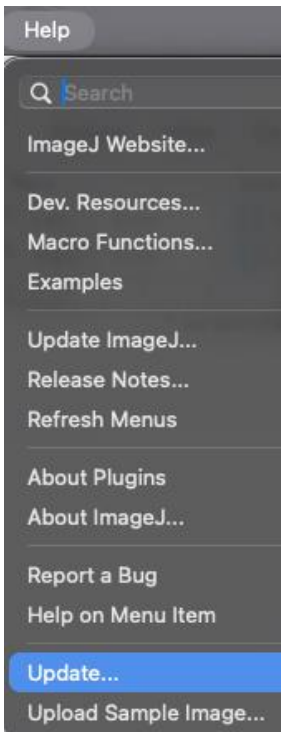
- Official and latest installation method can be found at https://fiji.sc/

Installing Wrappers for FIJI (ijl-utilities-wrappers) plugin

- Official and latest installation method can be found at https://github.com/BIOP/ijl-utilities-wrappers

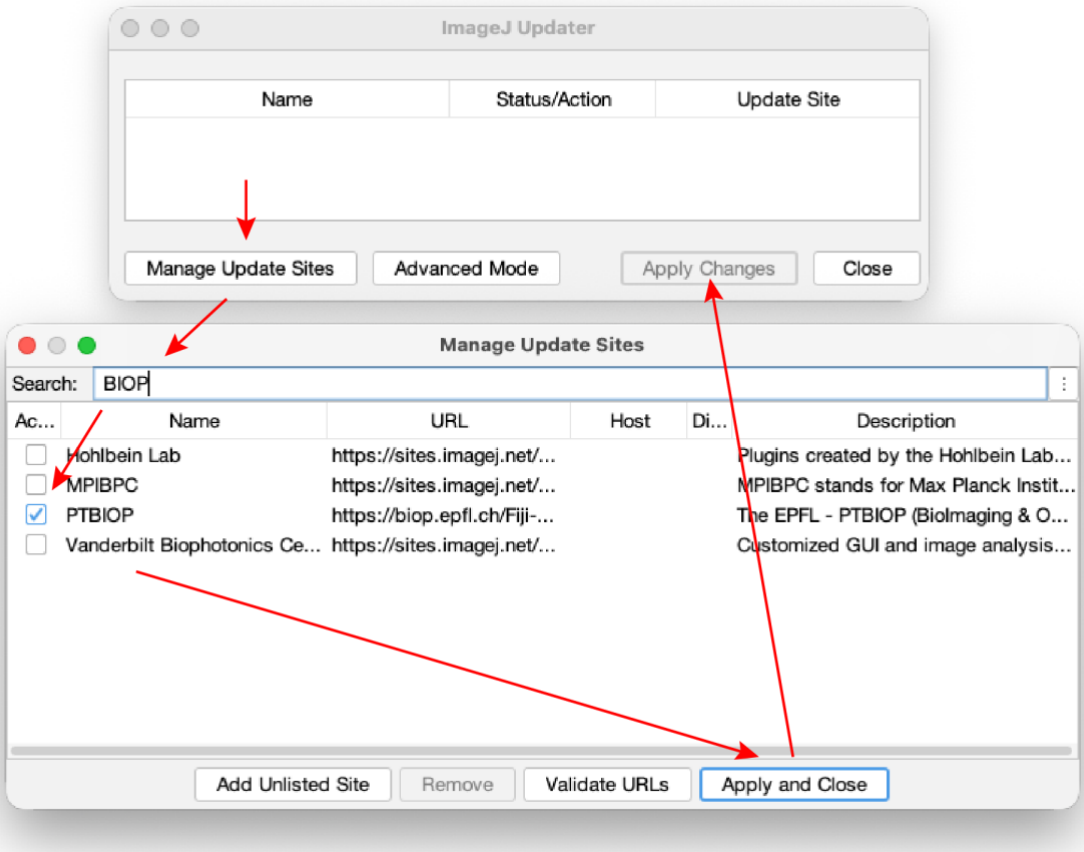
Start FIJI.

Select **'Help'** – **'Update…'**



Select **'Manage Update Sites'** – Search **'BIOP'** – check **'PTBIOP'**

– Click **'Apply and Close'** – **'Apply Changes'.**



Installing cellpose (cellpose 4)

- Official and latest installation method can be found at https://github.com/mouseland/cellpose

Open a terminal window (Command prompt or Terminal)

1. Create a new conda environment:

   ```
   conda create --name cellpose python=3.12
   ```

   - The environment name (cellpose) can be customised
2. Activate conda environment:

   ```
   conda activate cellpose
   ```

3. Install cellpose 4 (tested upto cellpose 4.0.4):

   ```
   pip install cellpose==4.0.4
   ```

4. Install cellpose GUI (graphical user interface):

   ```
   python -m pip install 'cellpose[gui]'
   ```

- Apple silicon may need to install pytorch to enable GPU support, found at https://pytorch.org/get-started/locally/

  install with: `pip3 install torch torchvision`

Install the 3DUptake macro

Download the latest version of the 3DUptake.zip from the GitHub page.

1. Download the 3DUptake.zip file from the release page

2. Extract the ZIP archive

3. Copy the 3DUptake folder into *Fiji installation folder*/scripts/Plugins

4. Restart Fiji to load the macros

The 3DUptake should now be accessible from the Fiji – Plugins – 3DUptake dropdown menu

4. Setup cellpose parameter

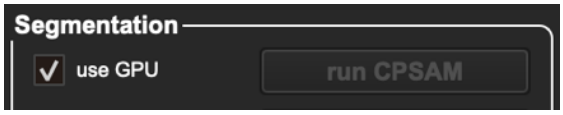
Once finished capturing data for analysis, optimise the Cellpose segmentation parameters directly using the GUI:

- Latest official instructions can be found at https://cellpose.readthedocs.io/en/latest/index.html

Open Terminal:

- `conda activate cellpose`
  (replace 'cellpose' with the environment name for cellpose– if renamed during installation)

python -m cellpose

- In cellpose GUI, on the left panel, under **Segmentation**, ensure a GPU is accessible to Cellpose. Note if:
  - 

 is checked
  - Command prompt displays:
    ```
    ** TORCH CUDA version installed and working. **
    ```
  - Terminal displays
    ```
    [INFO] ** TORCH MPS version installed and working. **
    ```
    (Apple silicon using MPS support only)
  - If GPU is not available, check the installation guide

Prepare sample image(s) for testing:

Open a test Z-stack image in Fiji

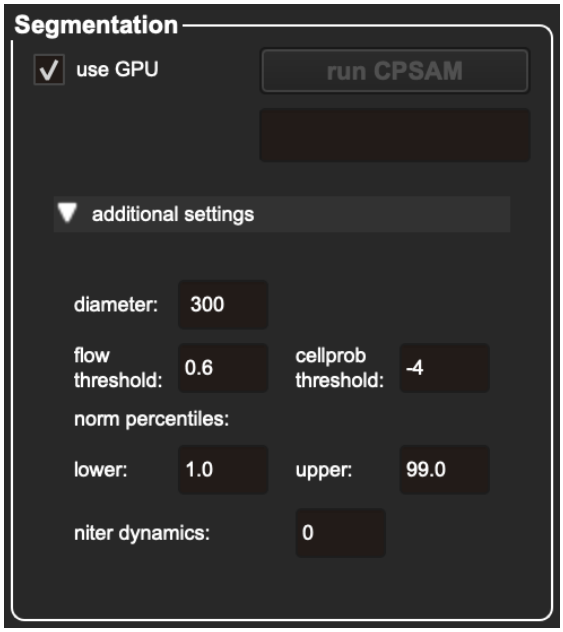
Plugins – 3DUptake – Generate Test Image

Save the "AVG_Test_Image" **AVG projection** as a separate sample file.

In the Cellpose GUI:

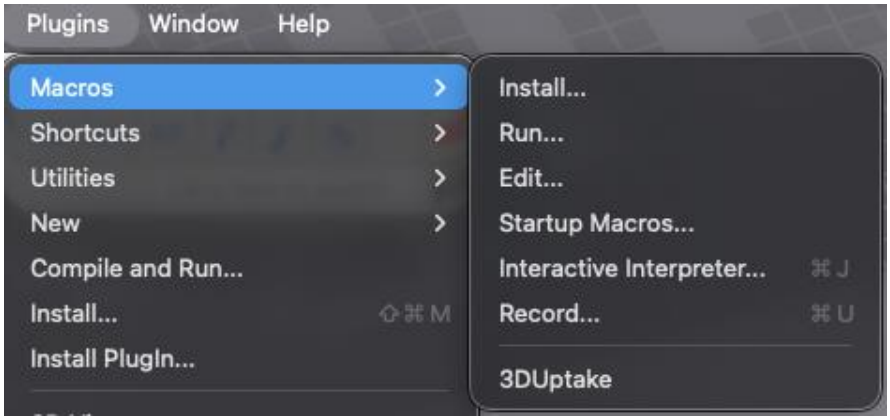
- Test both the 3D and SUM projected sample images by selecting:
  File – Load image (*.tif, *.png, *.jpg)

In the left panel, under **Segmentation**, Expand the “additional settings” tab, start testing with diameter at 200-300, flow threshold at 0.6 and cellprob threshold at -2 to -4. Leave the rest as default.

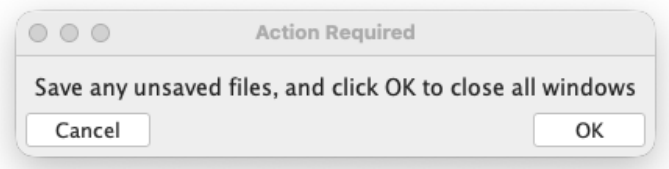


- Note the best settings for the specific imaging.
  - First cellpose run after installation automatically downloads the segmentation model, it will take time to load depends on the internet connection speed.
  - Test both the **SUM projection** sample file and **original** sample image. Depends on the cell type, the original image may produce better segmentation.

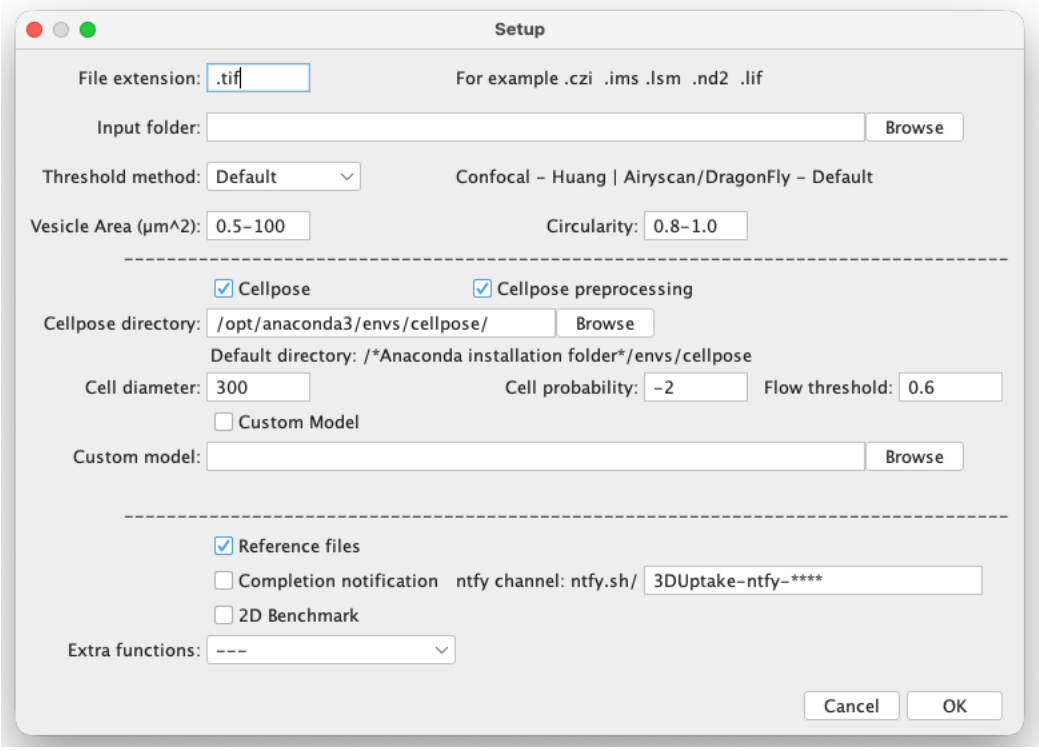
5. Running the 3DUptake Fiji Macro



- Choose Fiji – Plugins – Macros – 3DUptake
- If there is any opened window, save the needed files, then click ‘OK’



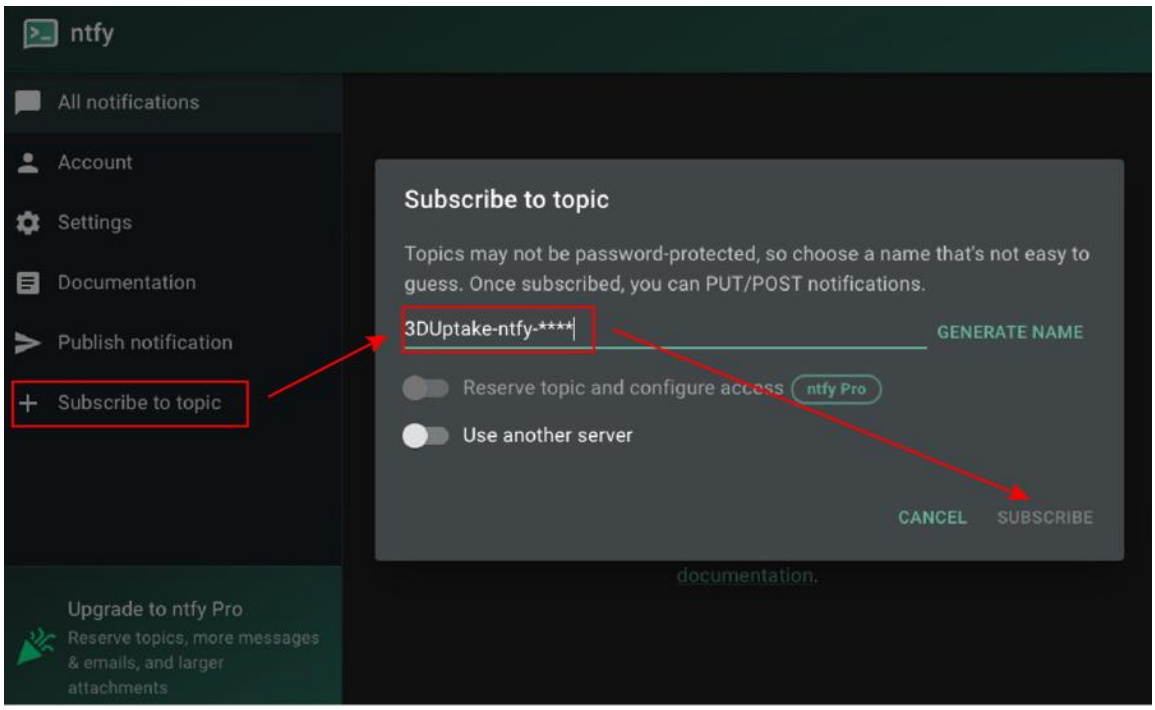
3DUptake Fiji Macro setup interface



- [**File extension**]: Input the image format. Non .tiff files will be opened using BioFormat importer.
  - Opening non .tiff file will be slower
  - BioFormat will run in windowless during the macro.
- [**Input folder**]: Click 'Browse' to choose the directory containing the dataset.
- [**Threshold method**]: Use 'Default' threshold method for initial testing, if the threshold is unsuitable for the dataset during quality check (refer chapter – quality check), the use of other method is supported. (Prioritise Huang if Default is unsuitable).
- [**Vesicle Area (µm^2)**]: Input the area size range for the object of interest. For macropinosomes, start with 0.5-100 (no upper limit)
- [**Circularity**]: Input the roundness filter for the target object. For circular vesicles, start with 0.8-1.0

- [**Cellpose**]: Use Cellpose for cell segmentation. Optional for analysis on the entire FoV.
- [**Cellpose preprocessing**]: Preprocess the images before Cellpose segmentation, optionally uncheck this option if the raw sample data produced better segmentation. For additional information, check 3. Optimising cellpose parameters.
- [**Cellpose directory**]: Input the directory for the cellpose environment. The default directory will be populated for both Windows and MacOS
- [**Cell diameter**], [**Cell probability**], [**Flow Threshold**]: Input the optimised cell diameter, cell probability, and flow threshold. Determine the optimised parameters using Cellpose GUI. For additional information, check 3. Optimising cellpose parameters.
- [**Custom Model**]: User can train their custom Cellpose model using the Cellpose GUI human-in-the-loop training. User can check this option and select the path to the custom model.

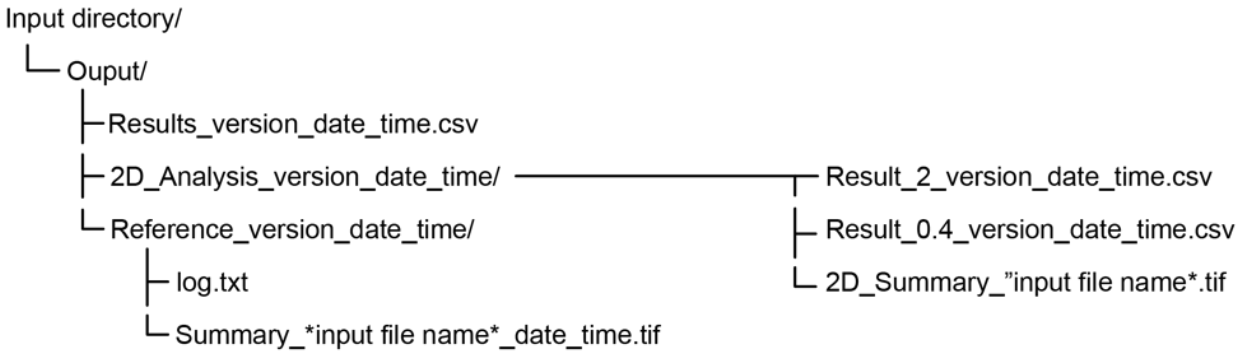
- [**Reference files**]: Strongly recommended. This option automatically produces visual quality check files.
  - Reference files are structured with CH1 (red): cell body, CH2 (green): vesicles, and CH3 (blue): cellpose segmentation.
  - Reference files are located in: /Input directory/References_*macro version*_*date*/
- [**Completion notification**]: Option to receive push notification when the macro finishes. Push notifications (mobile app or browser notification) are sent through open-sourced service ntfy.
  - Details can be found at: https://ntfy.sh/
  - Subscribe to the default ntfy channel 3DUptake-ntfy-**** (4 digits code will be randomly generated the first time this macro is started on each FIJI copy. Same FIJI installation will reuse the same channel on subsequent runs.
  - User can choose their own ntfy channel duing the setup and the channel will be saved for subsequent analysis on the same Fiji build.



- [**2D Benchmark**]: Comparing 3D and 2Danalysis. This feature needs to be used with cellpose enabled. When enabled, the macro will extract the second slice and slice at 40% of the stack (representing bottom and middle of the cells), before the slices are SUM projected and measured in 3D. The single slice will be analysed with the same method as the 3D and generating comparable spreadsheets and reference files in the output folder.
- [**Extra functions**]: These functions apply to edge cases unique to the systems used to develop this macro.
    - “Airyscan border removal” will remove black borders generated by Zen Black during Airyscan processing.
    - “Dragonfly de-noise” removes the bottom-most and 2 top-most slices from the input dataset. This function was specifically designed if the threshold of the input is incorrect due to high background – see troubleshoot

Click [**OK**] once the setup is finished. The log windows will be updated with a progress bar, and the notification and finishing prompt will be sent once the macro is finished.

6. Output structure



- Result_version_date_time.csv – primary result table
- Log.txt – Summary of run parameters
- Summary_*input file name*_date_time.tif – Quality check images
- Result_low_version_date_time.csv -2D analysis of the second slice of the 3D stack
- Result_min_version_date_time.csv -2D analysis of the slice at 40% of the 3D stack
- 2D_Summary_*input file name.tif – Quality check images for 2D benchmark

7. Key considerations

1. Signal to noise ratio – The SNR calculation for image captured with shadow imaging technique depends on which part of the image is considered ‘background’. If the dark cell body shadow is considered as the background, the SNR will need to be more than 20. Alternatively, if the fluorescent medium is considered as the background, it will typically need a SNR of 15 for successful and consistent threshold segmentation in Fiji.

2. Cell density – smaller cells and more cells in a field make it easier to acquire larger dataset for the experiment. With the latest version of cellpose 4 with cpsam model, densely packed adhering cells can be segmented cleanly. For our testing we seeded the cells at 0.1 M mL^{-1} and the RAW264.7 macrophages doubles overnight, and the final density is around 50 cells per FOV (40 after removing cells touching the boundary). The Andor Dragonfly system we used in this study has a FOV of 208 x 208 μm^2 when using a 60x objective and 1x camera magnification, a FOV with a fully confluent adherent cell layer of more than 100 cells can be segmented.

3. Acquisition speed – The acquisition for each stack should be kept as fast as possible. We highly recommend high-speed oriented system such as a spinning-disc confocal microscope for this workflow. Alternatively, other fast acquisition techniques such as fast Airyscan, or using resonant scanners are compatible as well. Using spinning-disc confocal microscopy each 3D image can be captured in around 5 s at Nyquist sampling rate (0.18 µm optical section thickness). However, optical section can be optimised to improve speed, see point 6.

4. Total acquisition duration – With high-speed microscope, a 3D image typically takes around 5 s to collect. Between each acquisition the user will need to find a new random position and adjust the Z coordinates for imaging. The time to locate and adjust for the new FOV determines the total number of FOVs that can be acquired within the uptake window. Therefore, the interval needs to be minimised. Generally, the interval between each 3D image was kept below 30 s to achieve a total of 20 acquisitions over 20 mins during early macropinocytosis.

To maximise the sample size within the set duration, the user can set multiple positions or use tiling for acquisition, however, we found multiple acquisition needs substantial amount of time during post-process to produce clean result. The specific interval between the first and last acquisition depends on the cell type. We recommend user to observe the endocytic traffic of cargo before set time interval for analysis.

5. Z-scanning range – When capturing dataset using shadow imaging technique, set the bottom of the cell at the Z-plane where the cell is **completely in focus**, out-of-focus slice greatly affect the threshold consistency. Similarly, set the top of the Z-stack before the of the cell disappears, see examples below. Do ensure that when setting the Z-scanning range to not include Z-planes containing only medium and no cell shadows.

6. Optical section thickness – Thinner optical sections can produce more detailed measurements, but compromise imaging time. With spinning-disc confocal microscope scanning at Nyquist sampling rate is generally not an issue. When using scanning confocal microscope, consider lowering the scanning resolution, using thicker optical section, or using lower pixel dwell time to maximise speed.

8. Quality check

Go to the reference file directory (/input directory/References_*version*_*date*)

Reference files are named as the original input, with “Summary_” prefix

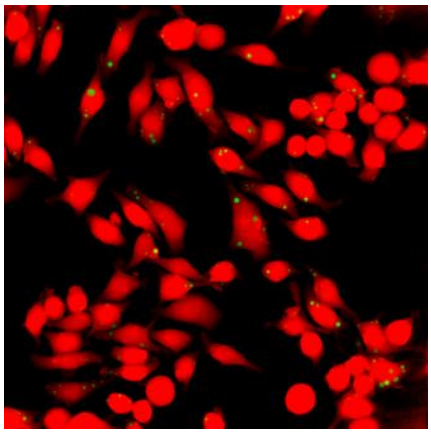
Open each file with FIJI

Check for

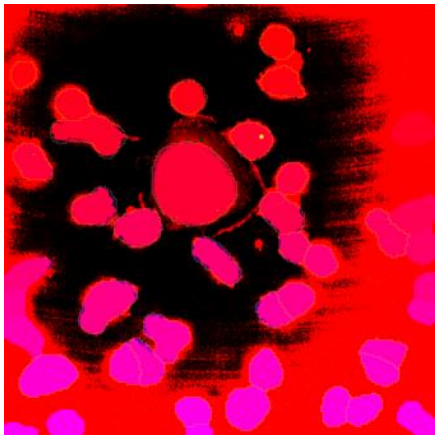
1. Threshold:

Correct threshold should preserve most of the cell boundary without high background.

Good threshold:



Background on the cell segmentation mask is too high:



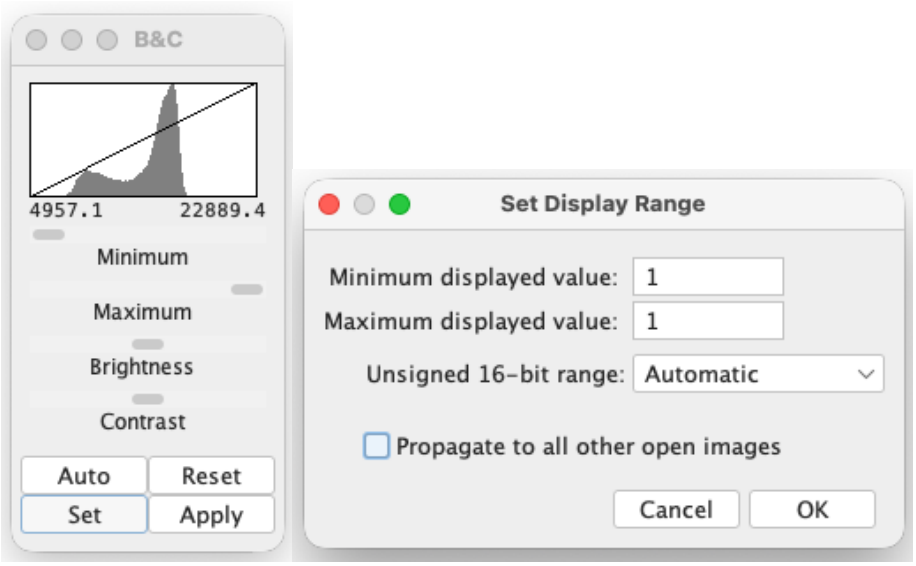
Check if the bottom of the Z-stack is in focus or optimises the threshold method

2. Noise

Correct vesicle segmentation should be free of noise

Set minimum and maximum Brightness/Contrast of Channel 2 (green) to 1

Inspect false positive vesicle mask

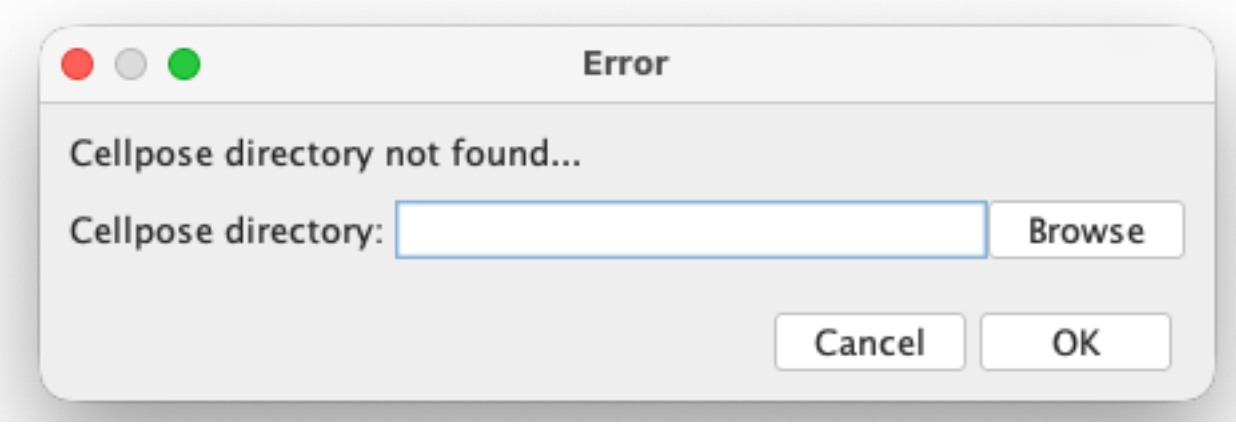


9. Troubleshoot

- Cellpose directory not found…

If the following window is displayed, the directory for cellpose is incorrectly configured.

This window will be persistent until a valid cellpose directory is selected.



Default Anaconda environment directory:

Windows: C:\Users\<your-username>\Anaconda3

MacOS: /opt/anaconda3/envs/*Cellpose environment name*

Linux: /opt/anaconda3/envs/*Cellpose environment name*

- Raise an issue ticket on GitHub page